Report R09 — Can R06's Finding Be Replicated?

Three attempts and a systematic screen of all 41 public lung cohorts. The answer is no, and the reason is not the biology.

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Summary

R06 reported that KEAP1 (HR 2.07) and SMARCA4 (HR 1.73) mutations are associated with shorter overall survival in NSCLC, and its own audit *withdrew* the accompanying replication claim: the second cohort it tried had 45 events in 604 patients and failed its own positive controls. That left the finding resting on a single cohort. This report is the attempt to fix that.

It cannot be fixed with public data, and the blocker is measurable. Three attempts:

1. **TCGA-LUAD + LUSC** — 978 patients, **385 events**, more than eight times the events of R06's failed attempt. Both positive controls still failed (TP53 HR 1.08, $q = 0.67$; STK11 HR 1.52, $q = 0.098$), in the full cohort *and* in a pre-declared late-stage subgroup. Median OS is 50.3 months against 28.8 in R06's cohort: this is resected disease, not advanced disease.

2. **MSK metastatic-organotropism LUAD** — rejected by a pre-registered setting gate before any hazard ratio was computed. Median OS 93.8 months, so despite the name it is not an advanced-disease cohort.

3. **A setting-matched pool** (NSCLC brain metastasis + a PDX-derived series, 380 patients, **231 events**, median OS 29.7 and 24.7 months). Median survival now matches R06's setting closely — and the controls fail again, with STK11 pointing the *wrong way* (HR 0.85 against R06's 1.63).

Between attempts, all 41 lung studies in cBioPortal were screened against five requirements fixed in advance. **Two qualified.** Seventeen have no overall-survival endpoint at all, fourteen have too few events, eight are the wrong disease setting.

What this means, stated carefully. This is not evidence that R06 is wrong. In every cohort tested, mutations whose prognostic effect in NSCLC is well established — TP53 and STK11 — were also undetectable, so no cohort demonstrated the competence needed to test the question. The honest conclusion is that **R06's finding remains a single-cohort result, and no public dataset can currently adjudicate it.** §5 specifies what would.

1 Attempt 1 — TCGA

Model was R06's, unchanged: per-gene unadjusted Cox on overall survival, Benjamini–Hochberg across the same five genes. Only patients in each study's `_sequenced` list were used; a patient without mutation profiling is not wild-type, and counting them as such would drag every hazard ratio toward 1.

Gene	Mutated	HR	q (BH)	R06's HR
KEAP1	140	1.02	0.88	2.07
SMARCA4	59	1.30	0.48	1.73
KMT2D	145	1.09	0.67	— (n.s.)
control TP53	658	1.08	0.67	1.25
control STK11	77	1.52	0.098	1.63

978 patients, 385 events, median OS 50.3 months.

STK11 is the near-miss worth noting: HR 1.52 against R06's 1.63 — almost the same effect size — but $q = 0.098$. TP53 is essentially absent. In the pre-declared late-stage (II–IV) subgroup (468 patients, 222 events), TP53 actually inverts to HR 0.79, which is biologically implausible and a clear signal that survival in this cohort is dominated by resection, adjuvant treatment and competing risks rather than by primary-tumour mutation status.

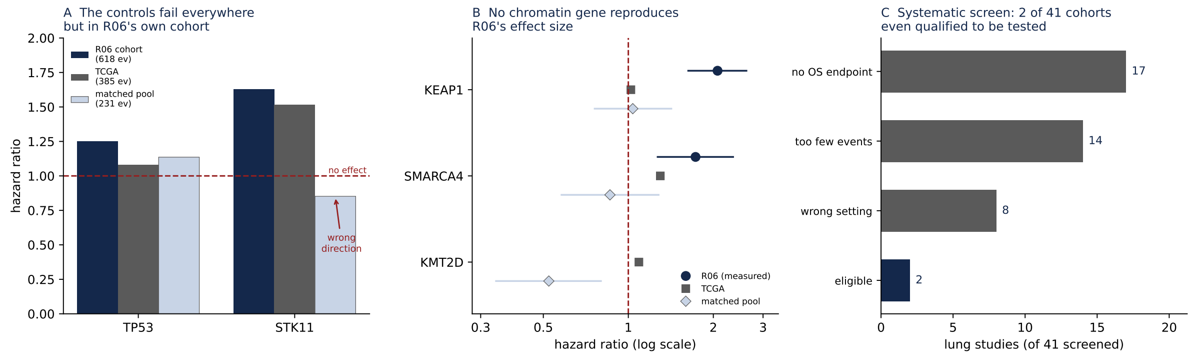
Because both controls failed in both strata, the pre-registered rule halted the run. That halt *is* the result: TCGA cannot test this question, despite having ample events.

2 The screen

Rather than stop after two failures, all 41 lung studies in cBioPortal were screened on requirements decided before any hazard ratio was computed: an OS endpoint; mutation profiling; at least 100 events; median OS nearer R06's 28.8 months than TCGA's 50.3; and patients disjoint from R06's cohort.

No overall-survival endpoint	17
Fewer than 100 events	14
Wrong disease setting (median OS too long)	8
Eligible	**2**
Total screened	41

The screen computes no outcome, only eligibility, so it cannot have been steered by results.



A The two positive controls in R06's cohort and in both attempts. Neither reproduces outside R06's own data, and STK11 inverts in the matched pool. **B** The three chromatin genes; R06's intervals are its measured 95% CIs. **C** Why there was nothing else to try.

3 Attempt 3 — the setting-matched pool

The two eligible cohorts: `bm_nsclc_mskcc_2023` (NSCLC brain metastasis, 209 patients, 114 events, median OS 29.7 months) and `lung_msk_pdx` (171 patients, 117 events, 24.7 months). Twenty-four overlapping patients were removed from the first before any outcome was read. Pooled analysis stratifies by cohort, so every comparison stays within-cohort and the two very different selections are never contrasted with each other.

Power was computed *before* running, so the control rule could not be rationalised afterwards: at 231 events STK11's HR 1.63 corresponds to $z \approx 2.6$ and should fire, while TP53's HR 1.25 gives $z \approx 1.7$ and may not. STK11 was therefore designated the load-bearing control.

Gene	Mutated	HR	95% CI	q (BH)
KEAP1	89	1.04	0.76–1.42	0.82
SMARCA4	49	0.86	0.58–1.28	0.58
KMT2D	52	0.52	0.34–0.80	**0.014**
control TP53	269	1.14	0.85–1.51	0.58
control STK11	66	0.85	0.59–1.23	0.58

STK11 came out at 0.85 — below 1, the opposite direction to R06's 1.63 — so the load-bearing control did not merely fail to reach significance, it pointed the wrong way. The run halted.

The KMT2D row is why that halt matters. It reaches $q = 0.014$, and in a report without a control gate it could have been written up as "KMT2D mutation is protective in advanced NSCLC". It is not interpretable: it appears in cohorts that cannot reproduce established biology, it contradicts R06 (which found KMT2D non-significant), it points in the protective direction, and R08 separately showed KMT2D's apparent methylation phenotype was pure histology confounding. Both of the strongest-looking KMT2D results in this series have dissolved on inspection.

4 What would actually settle it

Concretely, a cohort with all five of:

- **Advanced or metastatic NSCLC**, median OS in the 20–35 month range. This is the binding constraint — most public cohorts are surgically resected series.
- **At least 250 deaths**, and preferably 400. Not merely 250 patients.
- **Mutation calls** covering KEAP1, SMARCA4, KMT2D, STK11 and TP53.
- **Patients disjoint** from `nsclc_ctdx_msk_2022`.
- **Ideally a different institution.** Both eligible cohorts here were MSK, as R06's was; disjoint patients are not institutional independence, and this report does not claim otherwise.

If the group holds or can access such a cohort, `evidence/m9_tcga_replication.py` runs against it essentially unchanged — the model, gates, controls and replication criterion are fixed and hashed.

A caution that cuts the other way. Across three settings spanning 616 events, TP53 and STK11 were undetectable, and STK11 inverted in one. That is a strong hint that mutation–prognosis associations in NSCLC are highly dependent on disease setting and cohort selection. R06's own result should be read with that in mind: it may be specific to ctDNA-detected metastatic disease as sampled at one institution, rather than a general property of NSCLC. R06 remains correctly reported as a single-cohort finding.

5 Reproducing this report

- TCGA attempt: `evidence/m9_tcga_replication.py`; trace in `evidence/tcga_*`, including `gates_tcga_FIRST_HALT.json`
- Screen: `evidence/screen_replication_cohorts.py` → `evidence/screen_results.json` (all 41 studies with per-study verdicts)
- Matched attempt: `evidence/m10_matched_replication.py`; trace in `evidence/matched_*`, plus `gates_matched_CANDIDATE_REJECTED.json` for the cohort the setting gate rejected

- Every number quoted above is consolidated in `evidence/consolidated.json`
- Figure: `python3 make_figure.py`
- Data source: cBioPortal public REST API; no local data dependency